

Lucas Kowe

ELECTRICAL ENGINEER & APPLIED MATHEMATICIAN

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Qualified Electronic Engineer and Data Analyst with over a decade of diverse, strong professional experience in management, applied operations, data science and project management.

EDUCATION

B.Eng. Electrical (Electronic), BSci Mathematics (Hons) - University of Technology 2025

Thesis: Balancing Cost and Design: Creating a Modular LED Art Sculpture & Developing a Tool for Optimal Component Selection - Focus: Electronic Design, Optimisation & Operations Research

AREA OF EXPERTISE

Electronics	Business Intelligence	Data Analytics
Digital Design	Operations Research	Risk Management
Project Management	IT	Creative Electronic Design

ACHIEVEMENTS

- Migration of several large-scale information systems to new platforms
- Quantitative Operations Management in a growing SME
- Impactful projects including LED Sculptures enjoyed by millions of people
- Diverse data analysis experience in academic professional settings and private sector
- Completion of Tertiary Education whilst working in relevant full-time positions

PROFESSIONAL EXPERIENCE

Optik Consultancy x Nokia, Sydney – Intern Team Lead Nov 2025 – Feb 2026

- Led Consultancy cohort of 8 inter-disciplinary engineers to supply Nokia with a proof of concept design to showcase various patented technologies
- A mobile robotics platform was retrofitted with 5G connectivity and proprietary 8K 360° camera system to automate site safety using YOLO machine learning to detect and report PPE usage. The project involved mechanical design, Power design including PoE and control of the system comprising of Nokia RXRM technology and MQTT robotic control.
- The 12-week placement included responsibilities in project management, budgeting and supervision of a technical team of graduates.

AFTRS, Sydney – Database Administrator Jun 2024 – Nov 2025

- Database and Education Systems: Facilitate ETL integrations across identity directories, authentication services, learning platforms and cloud services. Databases ranged from 500 to 2000+ records.
- Infrastructure Development: Facilitated upgrades, development of automations and procedures enabling management and processes to be streamlined in the national school of 3000 students.
- General Technical Support: Provision of technical assistance to Academic staff and students.

Freelance, Sydney – Technical Consultant Mar 2023 - Present

- Web Development: Led the design and implementation of web content, collaborating with academics, stakeholders and webmasters to host digital creative content and track key analytics for use in research.
- Design Work (Vector Graphics and Publishing): Managed the production of ten print-ready and ten digital books for the Wonnarua Nation Aboriginal Corporation, contributing technical assistance.

Noble Works Australia, Sydney – Operations Manager Feb 2017 - Mar 2023

- Business Intelligence: Developed a costings model in Power BI to assist senior management in making informed business decisions on a project-by-project basis. 300+ Projects tracked ranging from small budget to longitudinal multi-million-dollar cost tracking.
- Operations Research: Provided insights to senior management aiding resultant company growth (up to 200% YoY) despite economic and competitive challenges.
- Operations: Oversaw day-to-day operations and managed expenditures against revenue.
- Web Development & Automation: Leveraged general programming skills and data science knowledge to create a website and automating processes to streamline operations reducing administrative costs and enhancing overall process efficiency.
- Compliance: Prepared demolition plans and required Health and Safety documentation for internal use, for provision to head contractors and large government and private clients, leading to project completion and positive feedback.
- Personal Development: Increased breadth of understanding of business processes in small, large and growing businesses, regulatory compliance and developed skills in practical projects.

Previous – ACU - Research Support, UWS – Research Support, Ord Minnett - Administrator

ADDITIONAL INFORMATION

- Programming Languages: Python, R, C/C#/C++, HTML, CSS, JS, PHP. SQL, M, DAX, R, Apps Script
- Spoken Languages: English – Native, Swedish - Conversational
- Certifications: PCEP-30-02 - Python Institute Python Programmer, UEERL0003 - Electrical Appliance Safety Testing, PUAEQU001 - Prepare, Maintain, Test Response Equipment, HLTAID011 - Provide First Aid
- Projects: Synaptic Array – Vivid 2017 Grant Winner, Various LED Sculptures, IllumiLux – Ambient lighting system
- Interests: Grappling, Diving, Travelling

REFEREES

Available upon request